Teaching Review: Addition of Angular Momenta

Concepts, Methods, and Worked Examples

Drawing from Griffiths (*Intro. to Elementary Particles*) and Zettili (*Quantum Mechanics*)

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Why Do We Need to Add Angular Momenta?

Before diving into the math, let us understand *why* this topic matters.

In quantum mechanics, angular momentum shows up everywhere. An electron in a hydrogen atom has **orbital** angular momentum $\mathbf{L}$ from its motion around the nucleus, and **spin** angular momentum $\mathbf{S}$ from its intrinsic quantum property. These are two *separate* angular momenta living in the same system. Nature does not keep them separate—they *couple* through the spin-orbit interaction, and the physically meaningful quantity is the **total** angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

Similarly, if you have two electrons (as in helium), each has its own spin. To describe the atom's total spin state you must *add* the two individual spins: $\mathbf{S}_{\text{total}} = \mathbf{S}_1 + \mathbf{S}_2$.

In particle physics (Griffiths), the same mathematics appears under a different name: **isospin**. A proton and neutron form an isospin-$\frac{1}{2}$ doublet. Pions form an isospin-1 triplet. When a pion scatters off a nucleon, you add isospin $1 \oplus \frac{1}{2}$, and the Clebsch–Gordan decomposition tells you the relative rates of different scattering channels.

The Central Question

Given two angular momentum quantum numbers j_1 and j_2, how do we describe the combined system? Specifically:

- What are the possible values of the **total** angular momentum quantum number j?
- How do we write the combined eigenstates $|j, m\rangle$ in terms of the individual states $|j_1, m_1\rangle\,|j_2, m_2\rangle$?

Review of Single Angular Momentum

Before adding two angular momenta, let us recall how *one* angular momentum works.

The Eigenvalue Equations

Any angular momentum operator $\hat{\mathbf{J}}$ satisfies the commutation relations:

$$[\hat{J}_x, \hat{J}_y] = i\hbar\,\hat{J}_z, \qquad [\hat{J}_y, \hat{J}_z] = i\hbar\,\hat{J}_x, \qquad [\hat{J}_z, \hat{J}_x] = i\hbar\,\hat{J}_y.$$

From these alone, one can derive that the simultaneous eigenstates of $\hat{J}^2$ and $\hat{J}_z$ satisfy:

$$\hat{J}^2\,|j, m\rangle = \hbar^2\, j(j+1)\;|j, m\rangle\,, \tag{1}$$

$$\hat{J}_z\,|j, m\rangle = \hbar\, m\;|j, m\rangle\,, \tag{2}$$

where $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ and $m = -j, -j+1, \ldots, j-1, j$.

For a given j, there are $2j+1$ values of m. So a spin-$\frac{1}{2}$ particle has 2 states, a spin-1 particle has 3 states, etc.

The Ladder Operators

The **raising** and **lowering** operators are defined as:

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y.$$

Their action is:

$$\hat{J}_\pm |j, m\rangle = \hbar\sqrt{j(j+1) - m(m \pm 1)}\ |j, m \pm 1\rangle .$$

These operators are the workhorses for constructing coupled states. They change m by ± 1 without changing j.

Physical Intuition for Ladder Operators

Think of the $2j + 1$ states as "rungs" of a ladder. $\hat{J}_+$ moves you up one rung (increases m), $\hat{J}_-$ moves you down one rung (decreases m). When you reach the top ($m = j$) or bottom ($m = -j$), the operator gives zero—you cannot go further.

Setting Up the Two-System Problem

Two Separate Hilbert Spaces

Suppose system 1 has angular momentum j_1 and system 2 has angular momentum j_2. System 1 lives in a $(2j_1 + 1)$-dimensional Hilbert space $\mathcal{H}_1$, and system 2 lives in a $(2j_2 + 1)$-dimensional space $\mathcal{H}_2$. The combined system lives in the **tensor product** space:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2,$$

which has dimension $(2j_1 + 1)(2j_2 + 1)$.

The Uncoupled Basis

The most natural basis for this product space uses the individual quantum numbers:

$$|j_1, m_1\rangle \otimes |j_2, m_2\rangle \ \equiv\ |j_1, m_1;\ j_2, m_2\rangle .$$

This is called the **uncoupled basis**. It consists of eigenstates of the set $\{\hat{J}_1^2,\ \hat{J}_{1z},\ \hat{J}_2^2,\ \hat{J}_{2z}\}$.

For example, if $j_1 = 1$ and $j_2 = \frac{1}{2}$, the uncoupled basis has $3 \times 2 = 6$ states:

$$|1, 1\rangle\,|\tfrac{1}{2}, \tfrac{1}{2}\rangle , \quad |1, 1\rangle\,|\tfrac{1}{2}, -\tfrac{1}{2}\rangle , \quad |1, 0\rangle\,|\tfrac{1}{2}, \tfrac{1}{2}\rangle , \quad |1, 0\rangle\,|\tfrac{1}{2}, -\tfrac{1}{2}\rangle , \quad |1, -1\rangle\,|\tfrac{1}{2}, \tfrac{1}{2}\rangle , \quad |1, -1\rangle\,|\tfrac{1}{2}, -\tfrac{1}{2}\rangle .$$

The Coupled Basis

Now define the **total** angular momentum:

$$\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2.$$

We can find simultaneous eigenstates of $\{\hat{J}^2,\ \hat{J}_z,\ \hat{J}_1^2,\ \hat{J}_2^2\}$. These are denoted $|j, m\rangle$ (with j_1 and j_2 understood). This is the **coupled basis**.

Common Misconception

Note that $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$, so the eigenvalue of $\hat{J}_z$ is always $m = m_1 + m_2$. But $\hat{J}^2 \neq \hat{J}_1^2 + \hat{J}_2^2$ (because of the cross term $2\hat{\mathbf{J}}_1 \cdot \hat{\mathbf{J}}_2$). So knowing j_1 and j_2 does **not** uniquely determine j—there are multiple possible values.

The Triangle Rule: What Values Can j Take?

The total angular momentum quantum number j can take the values:

$$\boxed{j = |j_1 - j_2|, \quad |j_1 - j_2| + 1, \quad \ldots, \quad j_1 + j_2 - 1, \quad j_1 + j_2.}$$

This is called the **triangle rule** (or **Clebsch–Gordan series**). In the notation of group theory, we write:

$$j_1 \otimes j_2 = |j_1 - j_2| \ \oplus \ (|j_1 - j_2| + 1) \ \oplus \ \cdots \ \oplus \ (j_1 + j_2).$$

Intuition for the Triangle Rule

Think classically: two vectors of length j_1 and j_2 can combine to produce a resultant of length anywhere from $|j_1 - j_2|$ (antiparallel) to $j_1 + j_2$ (parallel). Quantum mechanics restricts the result to integer steps within this range.
Also, the dimensions must match:

$$(2j_1 + 1)(2j_2 + 1) = \sum_{j=|j_1-j_2|}^{j_1+j_2} (2j + 1).$$

This is a valuable sanity check!

The Clebsch–Gordan Coefficients

Definition

The coupled and uncoupled bases are both complete—they span the same space. So we can expand one in terms of the other:

$$\boxed{|j, m\rangle = \sum_{\substack{m_1, m_2 \\ m_1 + m_2 = m}} \langle j_1, m_1;\ j_2, m_2 \,|\, j, m\rangle \ |j_1, m_1;\ j_2, m_2\rangle \,.}$$

The coefficients $\langle j_1, m_1;\ j_2, m_2 \,|\, j, m\rangle$ are called the **Clebsch–Gordan (CG) coefficients**. They are real numbers (by convention) that form a unitary matrix connecting the two bases.

Properties of CG Coefficients

Important Properties

1. **Selection rule:** $\langle j_1, m_1;\ j_2, m_2 | j, m\rangle = 0$ unless $m_1 + m_2 = m$.
2. **Triangle condition:** The coefficient vanishes unless $|j_1 - j_2| \le j \le j_1 + j_2$.
3. **Orthonormality:**
$$\sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | j, m\rangle \, \langle j_1, m_1; j_2, m_2 | j', m'\rangle = \delta_{jj'}\delta_{mm'}.$$
4. **Convention:** $\langle j_1, j_1;\ j_2, j_2 \,|\, j_1 + j_2,\ j_1 + j_2\rangle = 1$ (the stretched state coefficient is positive and equal to 1).

5. **Symmetry:** $\langle j_1, m_1; j_2, m_2 | j, m \rangle = (-1)^{j_1+j_2-j} \langle j_2, m_2; j_1, m_1 | j, m \rangle$.

How to Compute Them: The Ladder Method

This is the standard technique taught in both Zettili and Griffiths:

Step 1: Start with the stretched state. The highest-m state in the highest-j multiplet is unique:

$$|j_1 + j_2,\ j_1 + j_2\rangle = |j_1, j_1\rangle\, |j_2, j_2\rangle\,.$$

There is only one way to get $m = j_1 + j_2$: both subsystems must be "maxed out."

Step 2: Apply $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$ to step down. Use the lowering operator on both sides of the equation. The left side uses

$$\hat{J}_-\, |j, m\rangle = \hbar\sqrt{j(j+1) - m(m-1)}\ \ |j, m-1\rangle\,,$$

and the right side uses $\hat{J}_{1-}$ acting on $|j_1, m_1\rangle$ and $\hat{J}_{2-}$ acting on $|j_2, m_2\rangle$ (they operate on different spaces). Equate both sides, and you obtain $|j_{\max},\ j_{\max} - 1\rangle$.

Step 3: Repeat to generate all $|j_{\max}, m\rangle$ for $m = j_{\max}, j_{\max} - 1, \ldots, -j_{\max}$.

Step 4: Use orthogonality to find the next multiplet. For the $m = j_{\max} - 1$ subspace, there are *two* uncoupled states. One linear combination is $|j_{\max},\ j_{\max} - 1\rangle$ (which you just found). The *orthogonal* combination must be $|j_{\max} - 1,\ j_{\max} - 1\rangle$—the top of the next multiplet.

Step 5: Repeat steps 2–4 for each lower j until all states are found.

Why Does Orthogonality Work?

At each value of m, the number of uncoupled basis states with $m_1 + m_2 = m$ is exactly equal to the number of coupled states $|j, m\rangle$ that have that same m. Once you have found some of these coupled states via the ladder, the remaining ones must be orthogonal to them (since $|j, m\rangle$ and $|j', m\rangle$ with $j \neq j'$ are eigenstates of the Hermitian operator $\hat{J}^2$ with different eigenvalues, so they are automatically orthogonal).

Worked Example 1: Two Spin-$\frac{1}{2}$ Particles

Example 1: $\frac{1}{2} \otimes \frac{1}{2}$ — The Simplest Case

This is the first example every textbook covers. It describes the spin state of two spin-$\frac{1}{2}$ particles: two electrons, an electron and a proton, two quarks in a meson, etc.

Setting Up

We have $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$. Using the shorthand $|\uparrow\rangle = |\frac{1}{2}, +\frac{1}{2}\rangle$ and $|\downarrow\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$, the four uncoupled basis states are:

$$|\uparrow\uparrow\rangle\,, \quad |\uparrow\downarrow\rangle\,, \quad |\downarrow\uparrow\rangle\,, \quad |\downarrow\downarrow\rangle\,.$$

Triangle rule: j ranges from $|\frac{1}{2} - \frac{1}{2}| = 0$ to $\frac{1}{2} + \frac{1}{2} = 1$, so:

$$\tfrac{1}{2} \otimes \tfrac{1}{2} = 1 \oplus 0.$$

That gives $3 + 1 = 4$ states. ✓

Building the States Step by Step

Step 1: The stretched state ($j = 1, m = 1$).
The maximum m is $m = \frac{1}{2} + \frac{1}{2} = 1$. There is only one way to achieve this:

$$|1, 1\rangle = |\uparrow\uparrow\rangle\,.$$

This is the state where both spins point "up." The CG coefficient is trivially 1.

Step 2: Apply $\hat{J}_-$ to find $|1, 0\rangle$.
We apply $\hat{J}_-$ to both sides of $|1, 1\rangle = |\uparrow\uparrow\rangle$.
Left-hand side:

$$\hat{J}_-\,|1, 1\rangle = \hbar\sqrt{1(2) - 1(0)}\;|1, 0\rangle = \hbar\sqrt{2}\;|1, 0\rangle\,.$$

Right-hand side: Since $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$ and these act on different particles:

$$\begin{aligned}(\hat{J}_{1-} + \hat{J}_{2-})\,|\uparrow\uparrow\rangle &= (\hat{J}_{1-}\,|\uparrow\rangle)\,|\uparrow\rangle + |\uparrow\rangle\,(\hat{J}_{2-}\,|\uparrow\rangle) \\ &= \hbar\,|\downarrow\rangle\,|\uparrow\rangle + \hbar\,|\uparrow\rangle\,|\downarrow\rangle \\ &= \hbar\big(|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle\big).\end{aligned}$$

Equate and solve:

$$\hbar\sqrt{2}\;|1, 0\rangle = \hbar\big(|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle\big) \quad \Longrightarrow \quad \boxed{|1, 0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\big).}$$

Note that this is a **symmetric** superposition. Neither particle is individually "up" or "down"—they are entangled.

Step 3: Apply $\hat{J}_-$ again to find $|1, -1\rangle$.
Applying $\hat{J}_-$ to $|1, 0\rangle$: LHS gives $\hbar\sqrt{2}\;|1, -1\rangle$.

On the RHS, apply $\hat{J}_{1-} + \hat{J}_{2-}$ to $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$:

$$\frac{1}{\sqrt{2}}\left[\hat{J}_{1-}|\uparrow\rangle|\downarrow\rangle + \hat{J}_{2-}|\uparrow\rangle|\downarrow\rangle + \hat{J}_{1-}|\downarrow\rangle|\uparrow\rangle + \hat{J}_{2-}|\downarrow\rangle|\uparrow\rangle\right]$$
$$= \frac{1}{\sqrt{2}}\left[\hbar|\downarrow\downarrow\rangle + 0 + 0 + \hbar|\downarrow\downarrow\rangle\right] = \frac{2\hbar}{\sqrt{2}}|\downarrow\downarrow\rangle = \hbar\sqrt{2}\,|\downarrow\downarrow\rangle\,.$$

(The zeros come from $\hat{J}_{2-}|\downarrow\rangle = 0$ and $\hat{J}_{1-}|\downarrow\rangle = 0$—you cannot lower past $m = -\frac{1}{2}$.)

So:

$$|1,-1\rangle = |\downarrow\downarrow\rangle\,.$$

Both spins point "down"—as expected for $m = -1$.

Step 4: Use orthogonality to find the singlet $|0,0\rangle$.

We have used 3 of our 4 basis states. The remaining coupled state is $|0,0\rangle$ (with $j = 0$, $m = 0$).

In the $m = 0$ subspace, there are two uncoupled states: $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. We already know that $|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ is the symmetric combination. The *orthogonal* state must be antisymmetric:

$$\boxed{|0,0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\big).}$$

Let us verify: $\langle 1,0|0,0\rangle = \frac{1}{2}(1\cdot 1 + 1\cdot(-1)) = 0$. ✓

Summary Table and CG Coefficients

Coupled $\lvert j,m\rangle$	Expansion	CG coefficients	Name
$\lvert 1,+1\rangle$	$\lvert\uparrow\uparrow\rangle$	$\langle\uparrow\uparrow\,\lvert 1,1\rangle = 1$	3*Triplet
$\lvert 1,\ 0\rangle$	$\frac{1}{\sqrt{2}}(\lvert\uparrow\downarrow\rangle + \lvert\downarrow\uparrow\rangle)$	$\langle\uparrow\downarrow\,\lvert 1,0\rangle = \langle\downarrow\uparrow\,\lvert 1,0\rangle = \frac{1}{\sqrt{2}}$	
$\lvert 1,-1\rangle$	$\lvert\downarrow\downarrow\rangle$	$\langle\downarrow\downarrow\,\lvert 1,-1\rangle = 1$	
$\lvert 0,\ 0\rangle$	$\frac{1}{\sqrt{2}}(\lvert\uparrow\downarrow\rangle - \lvert\downarrow\uparrow\rangle)$	$\langle\uparrow\downarrow\,\lvert 0,0\rangle = +\frac{1}{\sqrt{2}}$, $\langle\downarrow\uparrow\,\lvert 0,0\rangle = -\frac{1}{\sqrt{2}}$	Singlet

Physical Meaning and Applications

What Does This Tell Us Physically?

The **triplet** ($j = 1$) has three states and is *symmetric* under particle exchange. It represents a state with net angular momentum $\hbar\sqrt{2}$. The **singlet** ($j = 0$) is *antisymmetric* and has *zero* total angular momentum—the two spins "cancel."

In atomic physics (Zettili): For a two-electron system (like helium), the Pauli exclusion principle requires the total wavefunction to be antisymmetric. If the spatial part is symmetric, the spin part must be the antisymmetric singlet—this is *parahelium*. If the spatial part is antisymmetric, the spin part is the symmetric triplet—this is *orthohelium*. The singlet–triplet structure directly affects the energy spectrum.

In particle physics (Griffiths): A quark-antiquark pair ($q\bar{q}$) with orbital angular momentum $L = 0$ can be in a spin singlet ($S = 0$) or triplet ($S = 1$). The pseudoscalar mesons (π, K, η)

are singlets; the vector mesons (ρ, K^*, ω, ϕ) are triplets. The CG decomposition is the reason these two families exist.

Worked Example 2: Spin-1 Plus Spin-$\frac{1}{2}$

Example 2: $1 \otimes \frac{1}{2}$ — Spin-Orbit Coupling and πN Scattering

This is the next level of complexity. In QM, it models spin-orbit coupling for a p-electron ($\ell = 1$, $s = \frac{1}{2}$). In particle physics, it models pion–nucleon isospin ($I_\pi = 1$, $I_N = \frac{1}{2}$).

Setting Up

Let $j_1 = 1$ (with states $|1\rangle, |0\rangle, |{-1}\rangle$) and $j_2 = \frac{1}{2}$ (with states $|\uparrow\rangle, |\downarrow\rangle$).
Triangle rule: j ranges from $|1 - \frac{1}{2}| = \frac{1}{2}$ to $1 + \frac{1}{2} = \frac{3}{2}$:

$$1 \otimes \tfrac{1}{2} = \tfrac{3}{2} \oplus \tfrac{1}{2}.$$

Dimension check: $3 \times 2 = 6$ and $(4) + (2) = 6$. ✓

The six uncoupled basis states are (writing $|m_1\rangle |m_2\rangle$):

$$\underbrace{|1\rangle|\uparrow\rangle}_{m=3/2}, \quad \underbrace{|1\rangle|\downarrow\rangle, |0\rangle|\uparrow\rangle}_{m=1/2}, \quad \underbrace{|0\rangle|\downarrow\rangle, |{-1}\rangle|\uparrow\rangle}_{m=-1/2}, \quad \underbrace{|{-1}\rangle|\downarrow\rangle}_{m=-3/2}.$$

Notice how we have grouped them by their total $m = m_1 + m_2$. The subspaces at the extremes ($m = \pm\frac{3}{2}$) have just one state each, while the interior subspaces ($m = \pm\frac{1}{2}$) each have two.

Constructing the $j = \frac{3}{2}$ Multiplet

Stretched state:

$$|\tfrac{3}{2}, \tfrac{3}{2}\rangle = |1\rangle|\uparrow\rangle.$$

Lower to $m = \frac{1}{2}$:

LHS: $\hat{J}_- |\frac{3}{2}, \frac{3}{2}\rangle = \hbar\sqrt{\frac{3}{2}\cdot\frac{5}{2} - \frac{3}{2}\cdot\frac{1}{2}}\ |\frac{3}{2}, \frac{1}{2}\rangle = \hbar\sqrt{3}\ |\frac{3}{2}, \frac{1}{2}\rangle$.
RHS:

$$\begin{aligned}(\hat{J}_{1-} + \hat{J}_{2-})|1\rangle|\uparrow\rangle &= (\hat{J}_{1-}|1\rangle)|\uparrow\rangle + |1\rangle(\hat{J}_{2-}|\uparrow\rangle) \\ &= \hbar\sqrt{1(2) - 1(0)}\ |0\rangle|\uparrow\rangle + \hbar\sqrt{\tfrac{1}{2}\cdot\tfrac{3}{2} - \tfrac{1}{2}(-\tfrac{1}{2})}\ |1\rangle|\downarrow\rangle \\ &= \hbar\sqrt{2}\ |0\rangle|\uparrow\rangle + \hbar\cdot 1\cdot|1\rangle|\downarrow\rangle.\end{aligned}$$

Equating: $\hbar\sqrt{3}\ |\frac{3}{2}, \frac{1}{2}\rangle = \hbar\sqrt{2}\ |0\rangle|\uparrow\rangle + \hbar\ |1\rangle|\downarrow\rangle$, so:

$$\boxed{|\tfrac{3}{2}, \tfrac{1}{2}\rangle = \sqrt{\frac{2}{3}}\ |0\rangle|\uparrow\rangle + \sqrt{\frac{1}{3}}\ |1\rangle|\downarrow\rangle.}$$

Reading off the CG coefficients:

$$\langle 1, 0;\ \tfrac{1}{2}, \tfrac{1}{2}\,|\,\tfrac{3}{2}, \tfrac{1}{2}\rangle = \sqrt{\frac{2}{3}}, \qquad \langle 1, 1;\ \tfrac{1}{2}, -\tfrac{1}{2}\,|\,\tfrac{3}{2}, \tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}}.$$

Lower to $m = -\frac{1}{2}$:

Applying $\hat{J}_-$ to $|\frac{3}{2},\frac{1}{2}\rangle$:
LHS: $\hat{J}_-\,|\frac{3}{2},\frac{1}{2}\rangle = \hbar\sqrt{\frac{3}{2}\cdot\frac{5}{2}-\frac{1}{2}(-\frac{1}{2})}\;|\frac{3}{2},-\frac{1}{2}\rangle = 2\hbar\;|\frac{3}{2},-\frac{1}{2}\rangle$.
RHS (applied to $\sqrt{\frac{2}{3}}\,|0\rangle\,|\uparrow\rangle + \sqrt{\frac{1}{3}}\,|1\rangle\,|\downarrow\rangle$):

$$\begin{aligned}
&\sqrt{\frac{2}{3}}\big[\hbar\sqrt{2}\;|-1\rangle\,|\uparrow\rangle + \hbar\;|0\rangle\,|\downarrow\rangle\big] + \sqrt{\frac{1}{3}}\big[\hbar\sqrt{2}\;|0\rangle\,|\downarrow\rangle + 0\big]\\
&= \hbar\sqrt{\frac{2}{3}}\cdot\sqrt{2}\;|-1\rangle\,|\uparrow\rangle + \hbar\Big(\sqrt{\frac{2}{3}}+\sqrt{\frac{2}{3}}\Big)\,|0\rangle\,|\downarrow\rangle\\
&= \frac{2\hbar}{\sqrt{3}}\;|-1\rangle\,|\uparrow\rangle + \frac{2\hbar\sqrt{2}}{\sqrt{3}}\;|0\rangle\,|\downarrow\rangle\,.
\end{aligned}$$

Dividing by $2\hbar$:

$$\boxed{|\tfrac{3}{2},-\tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}}\;|-1\rangle\,|\uparrow\rangle + \sqrt{\frac{2}{3}}\;|0\rangle\,|\downarrow\rangle\,.}$$

Bottom state:

$$|\tfrac{3}{2},-\tfrac{3}{2}\rangle = |-1\rangle\,|\downarrow\rangle\,.$$

Constructing the $j=\frac{1}{2}$ Multiplet by Orthogonality

In the $m=\frac{1}{2}$ subspace, we have two uncoupled states: $|0\rangle\,|\uparrow\rangle$ and $|1\rangle\,|\downarrow\rangle$. We already found:

$$|\tfrac{3}{2},\tfrac{1}{2}\rangle = \sqrt{\tfrac{2}{3}}\;|0\rangle\,|\uparrow\rangle + \sqrt{\tfrac{1}{3}}\;|1\rangle\,|\downarrow\rangle\,.$$

The orthogonal state is $|\frac{1}{2},\frac{1}{2}\rangle$. To construct it, swap the coefficients and flip one sign:

$$\boxed{|\tfrac{1}{2},\tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}}\;|0\rangle\,|\uparrow\rangle - \sqrt{\frac{2}{3}}\;|1\rangle\,|\downarrow\rangle\,.}$$

Check: $\langle\frac{3}{2},\frac{1}{2}|\frac{1}{2},\frac{1}{2}\rangle = \sqrt{\frac{2}{3}}\cdot\sqrt{\frac{1}{3}} + \sqrt{\frac{1}{3}}\cdot(-\sqrt{\frac{2}{3}}) = 0$. ✓
Similarly, in the $m=-\frac{1}{2}$ subspace:

$$\boxed{|\tfrac{1}{2},-\tfrac{1}{2}\rangle = \sqrt{\frac{2}{3}}\;|-1\rangle\,|\uparrow\rangle - \sqrt{\frac{1}{3}}\;|0\rangle\,|\downarrow\rangle\,.}$$

Physical Applications

Why This Matters

Atomic physics (Zettili, Ch. 7): For a p-electron ($\ell=1$) with spin $s=\frac{1}{2}$, the total angular momentum is $j=\frac{3}{2}$ or $j=\frac{1}{2}$. The spin-orbit Hamiltonian $H_{\text{SO}} \propto \mathbf{L}\cdot\mathbf{S}$ splits these two levels—this is the origin of the fine structure doublet in alkali atoms. The $j=\frac{3}{2}$ level lies higher (for less than half-filled shells), and the splitting is proportional to $\langle\mathbf{L}\cdot\mathbf{S}\rangle = \frac{1}{2}[j(j+1)-\ell(\ell+1)-s(s+1)]\hbar^2$.
The CG coefficients tell you the *probability* of measuring a particular m_ℓ, m_s combination when the system is in a definite j, m_j state. For example, if the electron is in $|j=\frac{3}{2}, m_j=\frac{1}{2}\rangle$, the

probability of finding $m_\ell = 0$ and $m_s = +\frac{1}{2}$ is $|\sqrt{2/3}|^2 = \frac{2}{3}$.
Particle physics (Griffiths, Ch. 4): The pion has isospin $I = 1$ and the nucleon has $I = \frac{1}{2}$. The Δ baryon resonance has $I = \frac{3}{2}$. The CG coefficients for $1 \otimes \frac{1}{2}$ directly give the **branching ratios** for Δ decay. For instance, $\Delta^+ \to \pi^0 p$ versus $\Delta^+ \to \pi^+ n$ involves the CG coefficients $\sqrt{2/3}$ and $\sqrt{1/3}$, predicting a $2:1$ ratio.

Worked Example 3: Two Spin-1 Particles

Example 3: $1 \otimes 1$ — A Nine-State Problem

This richer example demonstrates all the features of the CG machinery at once. It arises when coupling two spin-1 particles, or when adding two p-wave orbital angular momenta. In nuclear physics, it is relevant to deuteron structure.

Setting Up

We have $j_1 = 1$, $j_2 = 1$. We write $|m_1, m_2\rangle$ for $|1, m_1;\ 1, m_2\rangle$.
Triangle rule: $j = 0,\ 1,\ 2$.

$$1 \otimes 1 = 2 \oplus 1 \oplus 0.$$

Dimension check: $3 \times 3 = 9$ and $5 + 3 + 1 = 9$. ✓

Let us organize the nine uncoupled states by m:

$m = m_1 + m_2$	Uncoupled states
$+2$	$\|1, 1\rangle$
$+1$	$\|1, 0\rangle\,,\ \|0, 1\rangle$
0	$\|1, -1\rangle\,,\ \|0, 0\rangle\,,\ \|-1, 1\rangle$
-1	$\|0, -1\rangle\,,\ \|-1, 0\rangle$
-2	$\|-1, -1\rangle$

Notice the structure: the $m = 0$ subspace is 3-dimensional—the largest. This is where the most interesting mixing happens.

The $j = 2$ Quintuplet

Stretched state: $|2, 2\rangle = |1, 1\rangle$.
Lower to $m = 1$:

LHS: $\hat{J}_-\,|2, 2\rangle = \hbar\sqrt{2 \cdot 3 - 2 \cdot 1}\ \ |2, 1\rangle = 2\hbar\ \ |2, 1\rangle$.
RHS: $(\hat{J}_{1-} + \hat{J}_{2-})\,|1, 1\rangle = \hbar\sqrt{2}\ \ |0, 1\rangle + \hbar\sqrt{2}\ \ |1, 0\rangle$.
So:

$$\boxed{|2, 1\rangle = \frac{1}{\sqrt{2}}\big(|1, 0\rangle + |0, 1\rangle\big).}$$

Note the **symmetric** combination—both terms appear with the same sign and coefficient.

Lower to $m = 0$:

LHS: $\hat{J}_-\,|2, 1\rangle = \hbar\sqrt{6}\ \ |2, 0\rangle$.
RHS: Apply $(\hat{J}_{1-} + \hat{J}_{2-})$ to $\frac{1}{\sqrt{2}}(|1, 0\rangle + |0, 1\rangle)$:

$$\frac{1}{\sqrt{2}}\Big[\underbrace{\hat{J}_{1-}\,|1, 0\rangle}_{\hbar\sqrt{2}\,|0,0\rangle}\big|\ \ \big\rangle + |1, \cdot\rangle\ \underbrace{\hat{J}_{2-}\,|0\rangle}_{=\hbar\sqrt{2}\,|-1\rangle}\ + \underbrace{\hat{J}_{1-}\,|0\rangle}_{\hbar\sqrt{2}\,|-1\rangle}\,|1\rangle + |0\rangle\,\underbrace{\hat{J}_{2-}\,|1\rangle}_{\hbar\sqrt{2}\,|0\rangle}\Big]$$

Let me write this out more carefully. Applying the operators term by term on $\frac{1}{\sqrt{2}}(|1,0\rangle + |0,1\rangle)$:

$$\frac{1}{\sqrt{2}}\big[\hbar\sqrt{2}\,|0,0\rangle + \hbar\sqrt{2}\,|1,-1\rangle + \hbar\sqrt{2}\,|-1,1\rangle + \hbar\sqrt{2}\,|0,0\rangle\big]$$
$$= \hbar\big[|1,-1\rangle + 2\,|0,0\rangle + |-1,1\rangle\big].$$

Equating with $\hbar\sqrt{6}\,|2,0\rangle$:

$$\boxed{|2,0\rangle = \frac{1}{\sqrt{6}}\big(|1,-1\rangle + 2\,|0,0\rangle + |-1,1\rangle\big).}$$

The remaining states $|2,-1\rangle$ and $|2,-2\rangle$ follow by symmetry (or further lowering):

$$|2,-1\rangle = \frac{1}{\sqrt{2}}(|-1,0\rangle + |0,-1\rangle), \qquad |2,-2\rangle = |-1,-1\rangle\,.$$

The $j = 1$ Triplet (by Orthogonality)

In the $m = 1$ subspace, $\{|1,0\rangle\,, |0,1\rangle\}$ spans a 2D space. We found:

$$|2,1\rangle = \frac{1}{\sqrt{2}}(|1,0\rangle + |0,1\rangle) \quad \text{(symmetric)}.$$

The orthogonal combination is:

$$\boxed{|1,1\rangle_{j=1} = \frac{1}{\sqrt{2}}\big(|1,0\rangle - |0,1\rangle\big) \quad \text{(antisymmetric)}.}$$

Lowering this gives:

$$|1,0\rangle_{j=1} = \frac{1}{\sqrt{2}}\big(|1,-1\rangle - |-1,1\rangle\big), \qquad |1,-1\rangle_{j=1} = \frac{1}{\sqrt{2}}\big(|0,-1\rangle - |-1,0\rangle\big).$$

The $j = 0$ Singlet (by Double Orthogonality)

In the $m = 0$ subspace, we have three uncoupled states: $|1,-1\rangle$, $|0,0\rangle$, $|-1,1\rangle$. We have already used two:

$$|2,0\rangle = \frac{1}{\sqrt{6}}(|1,-1\rangle + 2\,|0,0\rangle + |-1,1\rangle),$$
$$|1,0\rangle_{j=1} = \frac{1}{\sqrt{2}}(|1,-1\rangle - |-1,1\rangle).$$

The third state $|0,0\rangle$ must be orthogonal to both. A short calculation (or Gram–Schmidt) gives:

$$\boxed{|0,0\rangle_{j=0} = \frac{1}{\sqrt{3}}\big(|1,-1\rangle - |0,0\rangle + |-1,1\rangle\big).}$$

Check orthogonality with $|2,0\rangle$*:* $\frac{1}{\sqrt{6}\cdot\sqrt{3}}(1\cdot 1 + 2\cdot(-1) + 1\cdot 1) = \frac{1}{\sqrt{18}}(1-2+1) = 0$. ✓
Check orthogonality with $|1,0\rangle$*:* $\frac{1}{\sqrt{2}\cdot\sqrt{3}}(1\cdot 1 + 0 + (-1)\cdot 1) = 0$. ✓

The Beautiful Symmetry Pattern

Exchange Symmetry under $m_1 \leftrightarrow m_2$

Look at the structure:

$$\begin{aligned} j = 2: &\quad \text{all states are } \textbf{symmetric} \text{ — e.g., } |1,0\rangle + |0,1\rangle \\ j = 1: &\quad \text{all states are } \textbf{antisymmetric} \text{ — e.g., } |1,0\rangle - |0,1\rangle \\ j = 0: &\quad \text{state is } \textbf{symmetric} \text{ — } |1,-1\rangle - |0,0\rangle + |-1,1\rangle \end{aligned}$$

(The $j = 0$ state looks like it has a minus sign, but under exchange $m_1 \leftrightarrow m_2$, the state $|1,-1\rangle \to |-1,1\rangle$ and vice versa, so it is indeed symmetric.)
General rule: For $j_1 = j_2$, the symmetry under particle exchange alternates: $j_{\max}$ is symmetric, $j_{\max} - 1$ is antisymmetric, and so on. Mathematically:

$$\langle j_2, m_2;\ j_1, m_1 \,|\, j, m\rangle = (-1)^{j_1+j_2-j} \langle j_1, m_1;\ j_2, m_2 \,|\, j, m\rangle .$$

For $j_1 = j_2 = 1$: $(-1)^{2-j}$ gives $+1$ for $j = 2, 0$ and -1 for $j = 1$.
This has profound physical consequences. For identical bosons (integer spin), the spatial wavefunction must be symmetric, restricting which j values are allowed. For identical fermions, it must be antisymmetric.

Physical Application

Where $1 \otimes 1$ Appears

In nuclear physics: The deuteron is a proton-neutron bound state. In the S-wave ($L = 0$) channel, the two nucleon spins ($s = \frac{1}{2}$ each) couple to $S = 1$ (the deuteron is a spin-1 particle). If we then consider the D-wave ($L = 2$) admixture and its coupling to $S = 1$, we encounter $1 \otimes 2$ CG coefficients. But for the tensor force acting within the $L = 1$ partial wave on two p-wave nucleons, $1 \otimes 1$ is directly relevant.
In particle physics (Griffiths): When classifying $q\bar{q}$ mesons with orbital angular momentum $L = 1$, the quark spins first couple to $S = 0$ or $S = 1$ (Example 1). Then $L = 1$ couples with $S = 1$ via $1 \otimes 1 = 2 \oplus 1 \oplus 0$, generating the J^{PC} quantum numbers of the P-wave mesons: the a_2 ($j = 2$), b_1 ($j = 1$), and a_0 ($j = 0$) states.

Summary: The Big Picture

What You Should Take Away

1. **The problem:** Given two angular momentum quantum numbers j_1 and j_2, express the coupled eigenstates $|j, m\rangle$ of total $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$ in terms of the uncoupled product states $|j_1, m_1\rangle\,|j_2, m_2\rangle$.

2. **Allowed values:** j ranges from $|j_1 - j_2|$ to $j_1 + j_2$ in integer steps. Always verify the dimension count $(2j_1 + 1)(2j_2 + 1) = \sum(2j + 1)$.

3. **The m rule:** $m = m_1 + m_2$ always. This is the easy quantum number—it just adds.

4. **The method:** Start with the stretched state (unique, CG = 1), use the lowering operator $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$ to generate the highest-j multiplet, then use orthogonality to find the next multiplet.

5. **CG coefficients** are the expansion coefficients. They are real, form a unitary matrix, and are tabulated in standard references. For small cases you can derive them by hand as shown above.

6. **Physical content:** $|\langle j_1, m_1;\, j_2, m_2 | j, m\rangle|^2$ is the **probability** of measuring individual quantum numbers m_1, m_2 when the system is in the coupled state $|j, m\rangle$ (or vice versa).

7. **Universality:** The same mathematical framework applies to spin, orbital angular momentum, isospin, and any other $SU(2)$ quantum number. Learn it once and use it everywhere.

Connecting Zettili and Griffiths

Zettili develops the theory from the ground up in Chapter 7 of his quantum mechanics text: the algebra of angular momentum, the tensor product structure, the derivation of CG coefficients via ladder operators, explicit tables, and applications to atomic spectroscopy (spin-orbit coupling, Zeeman effect, hyperfine structure).
Griffiths (in *Introduction to Elementary Particles*, Chapters 4–5) takes the CG machinery and applies it to **isospin** and **flavor symmetry** in particle physics. The notation changes (I instead of j, "proton/neutron" instead of "spin up/down"), but the mathematics is identical. The CG coefficients predict:

- How hadrons are classified into isospin multiplets,
- The relative rates of strong decay channels (branching ratios),
- Selection rules for which reactions are allowed by isospin conservation.

The deep lesson: angular momentum addition is not just a technique—it is a reflection of the $SU(2)$ symmetry structure that underlies both the rotation group and isospin symmetry. Mastering the CG decomposition gives you a tool that works across all of physics.